Week 5 Lecture Activity

Week 5 Lecture Activity: Exploring Relationships with Linear Regression

Ecological Context

Forest edges often have higher light availability, which can influence plant growth. You'll investigate whether **light availability** (PAR) is correlated with **sapling height**, and use regression to model this relationship.

PAR stands for "photosynthetically active radiation", and it is the wavelengths of light from 400-700nm that drive photosynthesis. In this case, its units are in micromoles per meter squared per second, which indicates the amount of photons received by a surface for a specific amount of time.

JASP Analysis Instructions

1. Download the sapling height and light data from here (https://umbc.box.com/s/ilj7d2fmfyantq2i603tj0o0so3je1ml).
2. Open JASP → Load the sapling height and light data.
3. Ensure your variables are the correct data type!
4. Go to Regression → Select Linear Regression. (Note: When viewing your results, keep an eye on M1, as this is the

model with both variables in it that we actually want to use.)

a. Set:
    i. **Dependent variable** = Height_cm
    ii. **Covariate (predictor)** = PAR_umol_m2s
b. Statistics → Coefficients → Select Estimates
c. Statistics → Display → Select Model fit and Regression equation
d. Plots → Residuals Plots → Select Residuals vs. predicted

5. Go to Descriptives → Descriptive Statistics
    e. Select for it to show the mean and standard deviation
    f. Customizable Plots → Scatterplot
        iii. Select "none" for graph above and to the right of plot
        iv. Show linear regression line and 95% confidence interval

6. Compare the "Coefficients" table to the "Regression Equations" table.
7. Compare your "Residuals vs. Predicted" plot to your scatterplot.
    g. (Optional) If you want more information on residual plots, this video (https://www.khanacademy.org/math/ap-statistics/bivariate-data-ap/xfb5d8e68:residuals/v/residual-plots) is a quick watch.

Questions to Answer / Deliverables:

- Upload a PDF (or otherwise) of your scatterplot of sapling height vs. light. (1 point)
- Report the slope, intercept, and R^2 of your linear regression. Interpret these values in ecological terms

(a.k.a., relating these values back to plant growth and light). (3 points)

- Does a strong correlation mean light causes increased height? (1 point)
- What other variables might influence sapling height in a forest? Give at least two ideas. (1 point)

This assignment will be due on Thursday, 10/01 at 11:59 PM.